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Microplastics and nanoplastics co-exposure modulates chromium bioaccumulation and physiological responses in rats

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ABSTRACT

The environmental fragmentation of plastics generates a mixture of plastic particles of various sizes, which frequently co-occur with other mobile and persistent environmental pollutants. Despite the prevalence of such scenarios, the interaction between micro- and nanoplastics (MNPs) and their combined effects with environmental pollutants, such as highly toxic hexavalent chromium (Cr(VI)), remain almost entirely unexplored in mammalian species. This study demonstrated that nanoplastic and microplastic particles co-aggregate and together influence Cr bioaccumulation patterns and related physiological alterations in rats. Following a four-week repeated intragastric exposure of Wistar rats to MNPs and Cr(VI), either alone or in combination, MNPs significantly enhanced Cr bioaccumulation in the liver, heart, brain, and skin. Under co-exposure conditions, Cr (VI) was the primary driver of cellular effects observed in the blood, including shifts in immune cell sub-populations (e.g., neutrophils, lymphocytes) and alterations in red blood cell indices, while serum biochemistry reflected limited physiological stress. MNPs *per se* decreased creatine kinase activity and increased cholesterol levels. In summary, polystyrene MNPs increase Cr(VI) distribution and bioavailability, but co-exposure does not uniformly exacerbate toxicity. Instead, their interaction may selectively alter physiological responses, emphasizing the need for a deeper understanding of their combined effects and potential health risks.

1. Introduction

Plastics are emerging pollutants that pose a threat to human and environmental health. It is estimated that the annual production of plastics is about 430 million tons, of which two thirds are waste after single use. Plastics production has grown exponentially due to 180 times increased consumption from 1950 to 2018 (Rafey and Siddiqui, 2021). Plastics are included in all spheres of life such as agriculture, transport, construction, electronics (Pan et al., 2020), or for medical purposes (Ilyas et al., 2018). While the versatility of plastics has revolutionized numerous industries and daily life, their increasing presence in the environment represents a largely unquantified but potentially large health challenge for various organisms including humans. Micro- and nanoplastics (MNPs), smaller plastic fragments originating from the

degradation/fragmentation of larger plastic objects or intentionally manufactured plastics at micro or nano scale, are of significant concern for human health (Campanale et al., 2020; Seewoo et al., 2023; Alijagic et al., 2024a).

Living organisms constantly ingest plastics, particularly MNPs, which can cause disturbances and injuries in the digestive tract, risk for translocation across epithelia, and far-reaching physiological alterations (Hurley et al., 2020; Seewoo et al., 2023). Moreover, MNPs can be further transported to organs via the bloodstream, with the limiting factor assumed at a capillary diameter of 5–8 µm. The retention time and fate of plastic particles in the bloodstream is currently unknown. Nevertheless, Leslie et al. (2022) showed that plastic particles, depending on their size, can be localized in immune cells, while some are attached to proteins, lipid particles or vascular endothelium. These

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findings raise concerns about the local and systemic health effects of MNPs (Ali et al., 2024). Increasing MNP exposure may worsen health impacts and contribute to the burden of non-communicable diseases (Krause et al., 2024). Most studies focus on uniform particle sizes (Alijagic et al., 2024a), but organisms are naturally exposed to mixed micro- and nanoplastics. Data on their combined toxicity remain limited, highlighting the need for studies on co-exposure scenarios to better understand their effects (Fontes et al., 2024).

Due to their large surface area, MNPs can adsorb and transfer environmental pollutants (Alijagic et al., 2024a). Heavy metals adsorbed on the surface of plastic particles have been found in many aquatic and terrestrial habitats (Vedolin et al., 2018; Turner et al., 2019). Compared to surrounding materials, MNPs exhibit concentrations of toxic metals and metalloids that are 10–100 times higher, highlighting their role as highly effective vectors (Acosta-Coley et al., 2019). Existing studies on fish, in which plastic particles were detected, suggest an increased uptake of heavy metals (Abidli et al., 2021; Sarkar et al., 2021). Sarkar et al. (2021) found an association between plastic particles and high levels of As, Cd, Cr, Cu, Ni, Pb, and Zn in water and fish meat. Assessment of the combined health risks of MNPs and heavy metals to human health is even more complex than MNPs themselves and the full dimensions of this health threat are yet unknown. It is believed that the combined action may increase the adverse effects and cause unpredictable damage to the biosphere (Liu et al., 2022). Also, the fate and causes of their adverse effects *in vivo* are yet unknown. For example, Cr (VI) is widely used in mining, metal and textile industries and in agriculture and chemical manufacturing, which results in unregulated disposal in terrestrial and aquatic environments. Cr(VI) can easily move through aquifers, contaminating different sources of drinking water and food and posing a severe threat to public health and the environment (Jobby et al., 2018; Tumolo et al., 2020). In cells, Cr(VI) is reduced to Cr (III) in the presence of ascorbate and glutathione, producing reactive oxygen species (ROS) and increasing the risk of oxidative stress and DNA double-strand breaks (Sobol and Schiestl, 2012; Clementino et al., 2018; Krawiec and Zhitkovich, 2023). Cr(VI) toxicity, even at low concentrations, is a significant health threat to various environmental species, including mammals (Suljević et al., 2023).

The co-occurrence of Cr(VI) and MNPs is likely in industrially impacted environments with Cr(VI) concentrations in contaminated water sources ranging from low $\mu\text{g/L}$ to several mg/L (WHO, 2020), MNPs have been detected at varying densities, from thousands to millions of particles per cubic meter (Koelmans et al., 2019). Despite their potential for combined exposure, studies investigating the co-pollution of Cr(VI) and MNPs remain limited. While physicochemical interactions between Cr(VI) and MNPs have been documented (Liao and Yang, 2020; You et al., 2023; Shao et al., 2023) and shown to cause adverse effects in aquatic organisms (Kim et al., 2022; Jeong et al., 2022), the role of MNPs as vectors for Cr(VI) and their associated physiological alterations in mammalian species remains largely unexplored. To our knowledge, this study is the first to investigate MNPs' role in Cr(VI) bioaccumulation patterns following digestive tract exposure in rats.

The overarching aim of this study was to assess individual and combined effects of MNPs and Cr(VI) in rats following the intragastric exposure by evaluating (1) changes in the body weight of the Wistar rats over the exposure period of four weeks, (2) the effects of MNPs on Cr bioaccumulation patterns in skin, brain, heart, liver, and kidney, (3) the impacts on hematological profile, including red and white blood cell indices, and (4) the impacts on serum biochemical profile, including kidney and liver function, basic lipid parameters, and serum proteins. We hypothesize that MNPs facilitate the transport and bioaccumulation of Cr(VI) in different organs, potentially altering its bioavailability and toxicity. This interaction may lead to distinct physiological responses compared to single exposures, with MNPs modulating Cr(VI)-induced effects rather than uniformly exacerbating toxicity.

2. Materials and methods

2.1. Animals and maintenance conditions

Forty 12-week-old Wistar rats with a body weight of 226.35 ± 14.07 g were bred in the Laboratory for Biochemistry and Physiology, Department of Biology, University of Sarajevo, under standard laboratory conditions ($t = 22^\circ\text{C}$, 12 h light/dark cycles). The rats were fed Oxbow's Essentials-Adult food, which is formulated to meet the nutritional needs of adult rats (crude protein 15 %, crude fat 4 %, crude fiber (min) 2 %, crude fiber (max) 5 %, moisture 10 %, calcium (min) 1 %, calcium (max) 1.5 %, phosphorus 0.8 %, vitamin A 8.000 IU/kg, vitamin E 125 IU/kg, vitamin A 8.000 IU/kg, vitamin E 125 IU).

2.2. Stock dispersions/solutions and particle characterization

Micro- and nano-sized polystyrene nanoparticles (MNPs) were purchased from Sigma Aldrich. Microplastics (cat. number 78462) had a diameter of 7 μm (standard deviation $< 0.2 \mu\text{m}$, coefficient of variation $< 2 \%$) and were supplied as an aqueous dispersion with a density of 1.05 g/cm^3 . Nanoplastics (cat. number 43302) had a diameter of 100 nm with a standard deviation of $\leq 0.01 \mu\text{m}$ and were also supplied as an aqueous dispersion. Particle mix was prepared by mixing equal volume/weight of micro- and nano-sized particles, maintaining the stock concentration at 100 mg/mL . Purchased stock particle dispersions were prepared in MilliQ water. For intragastric administration, particles were diluted in tap water to reach 1.4 mg/mL . MNPs size and shape was characterized by the means of field emission scanning electron microscopy (FE-SEM), using the SE-detector in an FE-SEM microscope (Zeiss Sigma 300 VP) at 15 kV. Before the analysis, MNPs were distributed on carbon adhesive and coated with Cr to achieve a good signal-to-noise ratio. Coating was done in a sputter coater (Quorum 150 T ES plus) and MNPs were coated with a 5 nm thick layer of Cr, as described by Alijagic et al. (2024b). In addition, micro- and nano-sized particles and their mixture were characterized in milliQ water by dynamic light scattering (DLS; Mastersizer 2000, Malvern Instruments, UK) to determine the particle size distribution and specific surface area.

2.3. Concentration rationale, rat exposure, and sample collection

Each animal was exposed to 1.4 mg of MNPs per week (corresponding to 200 μg of MNPs per day). MNPs at a concentration of 2 mg/kg causes disruption of the intestinal barrier *in vivo* (Qiao et al., 2021). Based on the average weight of the rats in this study (226.35 g), the daily intake of 200 μg falls within an acceptable and comparable range for human ingestion (Wang et al., 2023). Concentration of the $\text{K}_2\text{Cr}_2\text{O}_7$ (Sigma-Aldrich, USA) stock solution prepared in distilled tap water and used for intragastric administration was set at 8 $\mu\text{g/mL}$. Based on the estimated total exposure and relative contribution of drinking-water, and according to the WHO Guidelines for drinking-water quality regarding the Cr concentrations, each animal was exposed to 8 μg per week (1.14 μg per day) of Cr(VI).

Before intragastric administration, both individual MNPs and Cr(VI), as well as their mixture prepared in milliQ water, were ultrasonicated at a frequency of 37 kHz for 10 min at 35°C (Elmasonic S 100 H Ultrasonic Cleaner, Germany) to ensure homogenous particle dispersion and avoid aggregation/agglomeration.

The exposure was conducted for 28 days, and the animals were divided into four groups ($N = 10$ each). The animals were exposed repeatedly once per week. The water was administered by oral gavage (18-gauge rounded/bulb-tipped gavage; IACUC Standard Procedure, 2023), as follows (1) control group – intragastric administration of 1 mL tap water; (2) MNPs – intragastric administration of 1 mL MNPs; (3) Cr (VI) – intragastric administration of 1 mL $\text{K}_2\text{Cr}_2\text{O}_7$; (4) MNPs + Cr(VI) – intragastric administration of 1 mL MNPs + Cr(VI).

The weight of the animals was measured at the beginning of the

experiment and at the beginning of each week before exposure. After 28 days of exposure, rats were anesthetized with Ketamine-Xylazine combination according to UBC (2023) (100 mg/kg + 20 mg/kg, i.p) for blood sampling and then euthanized using a double dose. Organs, including liver, heart, brain, kidney, small intestine, and skin were collected and stored at -24°C for further analyses.

This work has received approval for research ethics (01/01-259-3-2024) from Faculty of Science, University of Sarajevo, Sarajevo, Bosnia and Herzegovina and a proof/certificate of approval is available upon request. Throughout the duration of the experiment, the standards and procedures described by the Universal Declaration of Animal Rights (UNESCO, 1978), Universal Declaration on Animal Welfare, and the Animal Protection and Welfare Law of Bosnia and Herzegovina (Official Gazette 25/09) were implemented.

2.4. Extraction and graphite furnace atomic absorption analysis

All glassware used in the extraction was previously soaked in a 10 % (v/v) nitric acid solution for 48 h to avoid cross-contamination. A microwave-assisted acid digestion method of all collected rat tissue samples was performed. For the extraction, $0.5\text{--}1.0 \pm 0.1$ mg of the collected and homogenized tissue samples was weighed and digested. Due to the complex nature of tissue samples, digestion was performed according to the slightly adjusted procedure (in order to reduce acid usage) given by Lee et al. (2022) using 7 mL of nitric acid (HNO_3 , 65 % suprapure, Merck, Germany) and a laboratory microwave system (Berghof MWS – 4 model, Germany). Upon completion of the digestion process, the digestate was diluted to 15 mL using ultrapure water with a resistance of $18 \text{ M}\Omega\cdot\text{cm}$ (Simplicity® Water Purification Systems, Merck KGaA, Germany).

A blank probe was prepared using the same digestion method to avoid the matrix effect without the sample/tissue. During data analysis, all values were corrected by subtracting the procedural blank. Chromium levels in the samples and blank were determined using a calibration curve based on standard solutions prepared from a single-element atomic stock solution of Cr (1,000 mg/L; Certipur grade, Merck, Germany), traceable to NIST (National Institute of Standards and Technology). Single element atomic stock solution of Cr (1 000 mg/L; Certipur grade, Merck, Germany) was used for preparing standard solutions for the calibration curve. Trace Cr analyses (total Cr) of tissue samples were performed by Graphite Furnace Atomic Absorption (GFAA) (Varian AA240Z). The obtained data are given as the mean $\pm$ standard deviation (SD) ($\mu\text{g Cr/kg}$) value. Additionally, spiking experiments were conducted for all tissues, except the brain due to insufficient weight, and recovery factors were calculated to verify the analytical quality control of the results.

2.5. Hematological profiling and red blood cell indices

A total of 0.5 mL of blood was collected in a test tube with ethylenediamine tetraacetic acid (EDTA). Cyanmethemoglobin Drabkin's method was used to analyze hemoglobin (Hb) concentration; 10 μL of blood and 2.5 mL of reagent were used. Red blood cell count (RBC) using Hayem's isotonic solution (10 μL blood + 1.99 mL solution) and white blood cell count (WBC) using Turk's solution (20 μL blood + 0.8 mL solution) were determined by counting blood cells in Bürker-Türk counting chamber (as explained in Suljević et al., 2023). Hematocrit (Hct) was determined by calculating the ratio of plasma and blood cells in a capillary tube after centrifugation for 10 min at 16 000 rpm (Hettich Haematokrit 24 centrifuge, Germany). RBC indices: MCV (mean corpuscular volume), MCH (mean corpuscular hemoglobin), and MCHC (mean cell corpuscular hemoglobin) were calculated from the ratio of Hb, Hct, and RBC count (see detailed description in Suljević et al., 2021).

The rest of the blood was used to prepare blood smears. Blood smears were prepared according to the modified Pappenheim method with Giemsa solution (Semikem, Bosnia and Herzegovina); instead of May-

Grünwald staining current study used methanol (p.a.) for fixation. After 30 min of staining, cells were analyzed using Olympus BX41 equipped with $100\times$ immersion objective. Differential WBC count (dWBC) was obtained by quantifying neutrophils (both segmented – Seg, and non-segmented – Non seg), lymphocytes (Lym), and monocytes (Mon). In addition, neutrophil/lymphocyte ratio (N/L ratio) was determined, as an emerging marker of low-grade inflammation (Buonacera et al., 2022).

2.6. Serum biochemical profiling

A total of 2 mL of blood without anticoagulants was collected for biochemical analysis. The blood was stored at room temperature for 30 min and centrifuged for 10 min at 2 500 rpm. Serum (0.5 mL) was separated from the blood after centrifugation. A total of 16 biochemical parameters were analyzed in the serum by the dry chemistry method using the Ortho-Clinical Diagnostics VITROS® 350 System (Ortho Clinical Diagnostics, USA). The potentiometric method for Cl^- (chloride) and Na^+ (sodium) used electrolyte reference fluid (ERF) to analyze the concentration of serum electrolytes. Other parameters were detected using the spectrophotometric principle as follows: CK – creatine kinase activity (CKMB Slides), LDH – Lactate dehydrogenase activity (LDHI, IFCC Reference Method), AST – aspartate aminotransferase activity, ALT – alanine aminotransferase activity, Cre – creatinine (by creatinine amidohydrolase), BUN – blood urea nitrogen (by urease), Bil – bilirubin (total bilirubin by dyphilline), Glu – glucose (by GOD-PAP method), Prot – protein (total protein, biuret method), Alb – albumin (BCG method), Glob – globulin (total protein-albumin), A/G ratio – albumin/globulin ratio, Chol – cholesterol (by cholesterol esterase) and Trig – triglycerides (by lipoprotein lipase). All analyses were performed using Ortho-Clinical Diagnostics VITROS® MicroSlides (UK). To ensure accuracy and reliability, the VITROS® 350 System was calibrated using the Ortho-Clinical Diagnostics VITROS® Calibrator Kit 3. Additionally, VITROS® Control Performance Verifier I Level 1, an assayed control, was used to monitor system performance.

2.7. Statistical analysis

Data analysis was carried out using GraphPad Prism 10.2.3 statistical software (USA). A one-way ANOVA followed by a post-hoc Dunnett's test was used to determine if the measured values in the exposed groups significantly differed from the control group. In all analyses, $*P < 0.05$ was considered statistically significant, $**P < 0.01$ very significant, and $***P < 0.001$ extremely significant.

3. Results and discussion

3.1. Nanoplastic and microplastic particles co-aggregate

To assess the size distribution, surface area, and shape of polystyrene micro- and nano-sized particles, and their mixture (MNPs), DLS and FE-SEM analyses were conducted. The SEM images of MNPs shown in Fig. 1A–B display micro-sized particles with an average diameter of 7 μm , a size aligning with the manufacturer's specifications and DLS analysis (Fig. 1E, blue bars). At higher magnification (Fig. 1C–D), smaller particles of about 100 nm were also observed, consistent with DLS results (Fig. 1E). Notably, two significant findings emerged. First, nano-sized polystyrene particles were adhering to the surface of micro-sized particles or co-aggregating, as observed in SEM images (Fig. 1D) and the peak shift in DLS results for micro-sized particles in the mixture (Fig. 1E, red bars). This substantially increased the overall surface area. For example, the specific surface area of the micro-sized particles, when analyzed individually, was $0.894 \text{ m}^2/\text{g}$, whereas, when the particle mixture was analyzed using DLS, the surface area increased to $17.4 \text{ m}^2/\text{g}$, representing a 19.5-fold increase. Second, nano-sized particles appeared to promote the aggregation of micro-sized particles by forming

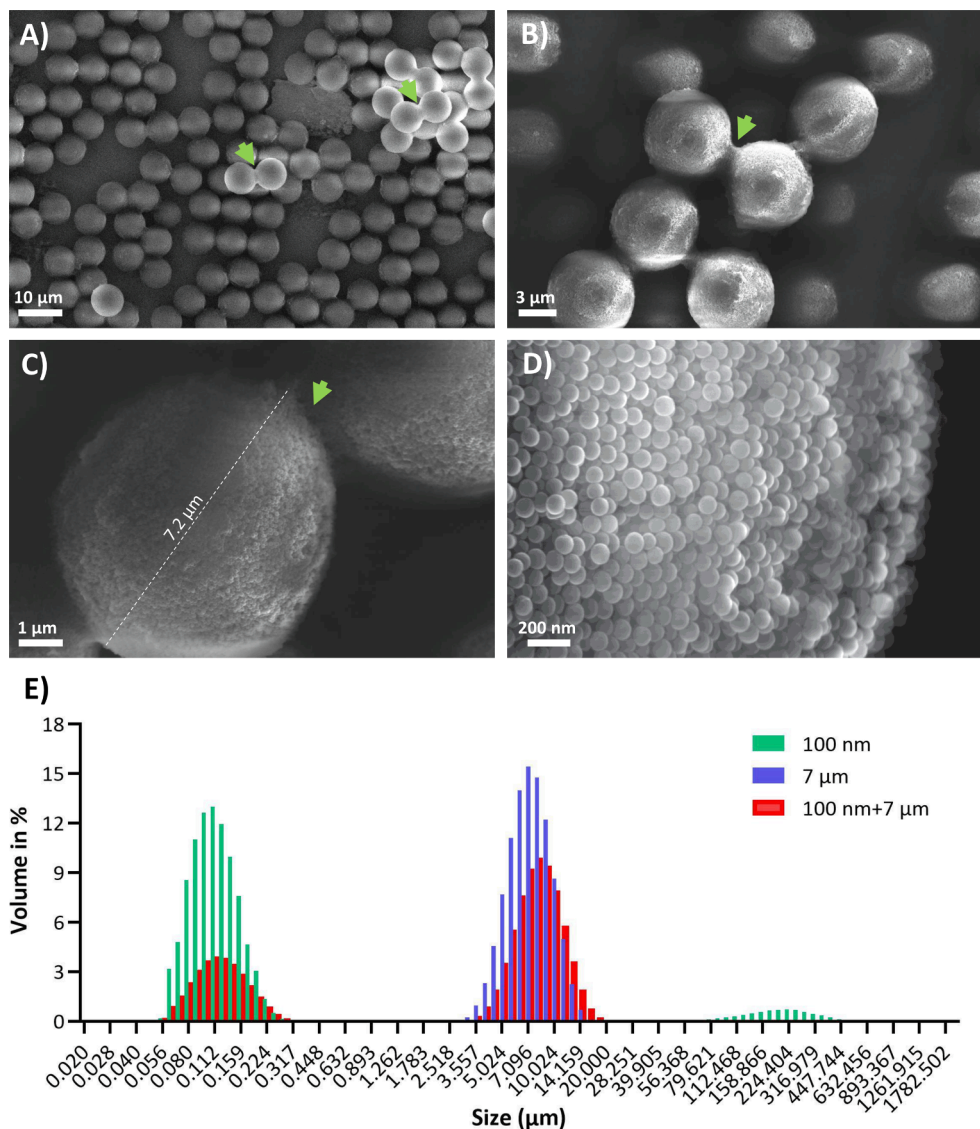


Fig. 1. Morphological and size evaluation of polystyrene micro- and nanoplastics (MNPs). (A–D) field-emission scanning electron microscopy (FE-SEM) of MNPs mixture. (E) Dynamic light scattering (DLS) analysis of microplastics (7 µm), nanoplastics (100 nm), and their mixture.

structural bridges between them (see Fig. 1A–C, green arrows). This is a particularly important observation, as plastic fragmentation under environmental conditions generates mixtures of both micro- and nano-sized particles, and there remains a significant knowledge gap regarding the interactions between these different size fractions. Altogether, these findings could have important implications for the design of MNP toxicity studies, both *in vivo* and *in vitro*. Nonetheless a fraction of the nano-sized polystyrene particles, as observed in DLS analysis, remained unbound to the micro-sized particles (Fig. 1E), potentially allowing also free nanoparticles to interact independently during the exposure.

While our study focused on morphological and size distribution analyses using FE-SEM and DLS, future studies should incorporate zeta potential measurements to better characterize surface charge properties and understand mechanisms governing the interaction between particles. Additionally, Derjaguin–Landau–Verwey–Overbeek (DLVO) theory provides a valuable framework for explaining the interplay between attractive and repulsive forces governing particle interactions (Ninham, 1999). By using extended DLVO, Wang et al. (2015) demonstrated that hydrophobic attractions were the dominant force between plastic particles.

A growing body of evidence suggests that MNPs act as vectors for both inorganic and organic chemicals (Brennecke et al., 2016; Sarkar et al., 2021; Zambrano-Pinto et al., 2023). Interestingly, Ho and Leung (2021) reported that heavy metals may have a crucial role on the interaction of engineered nanoparticles with polystyrene microplastics. They facilitate the formation of microplastic-heavy metal-nanoparticle complexes through cation bridging, charge neutralization, and surface chemistry modifications. Subsequent release of microplastics-adsorbed nanoparticles and heavy metals varied significantly between aquatic media and different simulated digestive fluids, confirming the important role of micro-sized plastics for transfer of nano-sized particles from, for example, natural environments into organisms (Ho and Leung, 2021). To the best of our knowledge, no studies have examined the interaction between micro-sized and nano-sized plastic particles, especially in the context of co-exposure with heavy metals. In a recent study, Shaoyong et al. (2024) demonstrated that Cr loading alters the toxicity profiles of polystyrene particles. To better understand the role of MNPs as vectors, including for heavy metals like a highly toxic Cr(VI), their effects on Cr bioaccumulation in vital tissues and physiological responses were examined by exposing rats to MNPs alone, Cr(VI) alone, and their combination, followed by a series of complementary analyzes in blood,

serum, and various tissues/organs.

3.2. Impact of MNPs and Cr(VI) (co-)exposure on animal body weight

To assess the effects of MNPs, Cr(VI), and MNPs + Cr(VI) exposures on body weight, animals were weighed at the beginning of the experiment and prior to each exposure throughout the four-week study period. Fig. 2 displays the average body weight of all animals. Limited increase was observed in the second week when comparing the weight of MNPs and Cr(VI) groups to the control group. This weight gain was unexpected, and while the mechanisms remain unclear, it may be influenced by factors such as alterations in gut microbiota, metabolic changes, or individual variability in physiological responses. Some studies suggest that exposure to MNPs can modulate lipid metabolism and energy balance (Wang et al., 2022; Luo et al., 2025), potentially contributing to fluctuations in body weight. Future studies should explore metabolic and nutritional parameters to clarify the underlying mechanisms. The other groups did not exhibit any significant changes in weight. Existing data on the effects of MNPs on body weight gain in exposed rats are conflicting; some studies report a decrease (Haddadi et al., 2022; Saeed et al., 2023), while others report no changes upon exposure (Kim et al., 2021).

3.3. MNPs increase chromium bioaccumulation in different organs

To investigate bioaccumulation patterns of total Cr under Cr(VI) exposure and co-exposure with MNPs, extraction followed by graphite furnace atomic absorption spectrometry was performed. The accuracy of the method was confirmed based on satisfactory recovery factors (95–105 %). The limit of detection for Cr content in tissues, determined as the mean blank value plus three standard deviations ($\bar{X}_{bl} + 3 \times SD_{bl}$, $n = 5$), was 0.39 $\mu\text{g/L}$. Total Cr was quantified as a large portion of Cr internalized by cells is reduced to Cr(III) (Ray, 2016; Clementino et al., 2018). Under Cr(VI) single exposure, significantly higher levels of total Cr were observed only in the skin of the exposed animals compared to the control animals (Fig. 3). In contrast, in other organs, including the heart, brain, and liver significantly higher ($P < 0.05$) bioaccumulation of Cr was observed when rats were co-exposed to MNPs and Cr(VI) (Fig. 3). These findings hint that MNPs may enhance the absorption or retention of Cr in various tissues, thereby increasing its bioaccumulation. Furthermore, MNPs acting as vectors for Cr distribution to organs distant

from the initial exposure site (i.e., digestive epithelia) may lead to more widespread physiological impacts under co-exposure conditions. This is particularly significant given that nano-sized particles were found adhered to the surface of micro-sized particles (Fig. 1), increasing their surface area manifold and thus providing more available surface for Cr adsorption and transfer to distal organs and target cells. While most MNPs are excreted via feces (Seewoo et al., 2023), a fraction that remains in the body may contribute to the long-term retention of Cr, potentially affecting its bioaccumulation and systemic distribution. The fate of Cr translocated by MNPs into internal tissues and organs is influenced by multiple factors. The desorption process from MNPs is highly complex and depends on various interacting variables, including the physical and chemical properties of MNPs, Cr, and the surrounding medium. The strength and number of mechanisms involved are directly linked to factors such as the Cr-MNP bond type (e.g., van der Waals forces, electrostatic interactions, hydrogen bonding, π - π interactions, and cation-ligand interactions), as well as pH, salinity, and temperature (Fred-Ahmadu et al., 2020). This phenomenon may further affect Cr mobility and bioavailability within the body (Hou et al., 2023). A previous study demonstrated that Cr-loaded polystyrene particles, approximately 200 nm in diameter, can cross the intestinal barrier and accumulate in various tissues, including the heart, liver, spleen, lungs, and kidneys. Furthermore, Cr loading on the particle surface enhances the transport of these particles across the intestinal barrier into organs (Shaoyong et al., 2024).

This study used a 1:1 (w/w) ratio of micro- and nano-sized polystyrene particles to assess their combined effects; however, different mixing ratios may influence key parameters such as particle aggregation, surface area exposure, and interactions with Cr, potentially altering bioaccumulation patterns and physiological responses. Future co-exposure studies should explore how varying proportions of micro- and nano-sized particles affect bioavailability and toxicity to further refine our understanding of their interactions.

Shaoyong et al. (2024) reported that Cr loading does not significantly alter the size or surface functional groups of aged polystyrene particles; however, it does modify their surface topography. In contrast, since the current study used pristine polystyrene MNPs, the absence of significant surface modifications upon Cr loading aligns with expectations for hydrophobic polymers lacking highly reactive functional groups. However, Shao et al. (2023) demonstrated that UV aging increases surface roughness and oxidation, leading to the formation of oxygen-containing functional groups that enhance Cr(VI) adsorption. This suggests that environmental aging may alter the interaction between MNPs and Cr, potentially influencing bioaccumulation and toxicity. Furthermore, Cr adsorption is polymer-specific, with variations in surface chemistry influencing metal binding affinity (Shao et al., 2023).

3.4. Assessment of baseline immunological status in response to MNPs and Cr(VI) (co-)exposure

In order to assess the baseline immunological status in response to MNPs and Cr(VI) single and combined exposure, the WBC and dWBC counts were assessed (Fig. 4). WBC values were significantly lower ($P < 0.05$) in all exposed groups (MNPs, Cr(VI), and MNP + Cr(VI)) compared to the control group. The lowest values were recorded in the Cr(VI) group, while no significant differences were observed between the MNPs and Cr(VI) + MNPs groups. Segmented neutrophils were also significantly lower in all exposed groups compared to the control. Notably, the MNPs-exposed rats displayed the lowest values of segmented neutrophils, while non-segmented neutrophils showed a shift, being the highest compared to all other groups (i.e., Cr(VI) and MNPs + Cr(VI)). Lymphocyte counts were significantly ($P < 0.05$) higher in all exposed groups compared to the control, with the highest values recorded in the MNPs group. Additionally, the N/L ratio was significantly ($P < 0.05$) lower under all exposure scenarios compared to the control. Monocyte levels were significantly elevated only in the Cr(VI) group.

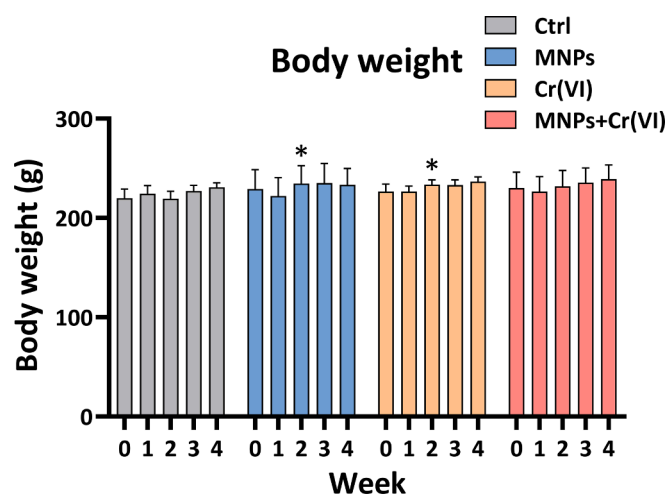


Fig. 2. The body weight of Wistar rats over the exposure to polystyrene micro- and nano-sized particles and chromium(VI). The data are reported as the mean $\pm$ standard deviation of 10 rats ($N = 10$) per exposure condition. Statistical significance was assessed by one-way ANOVA followed by the post hoc Dunnett's test. * $P < 0.05$. Micro-nanoplastics – MNPs; chromium(VI) – Cr(VI).

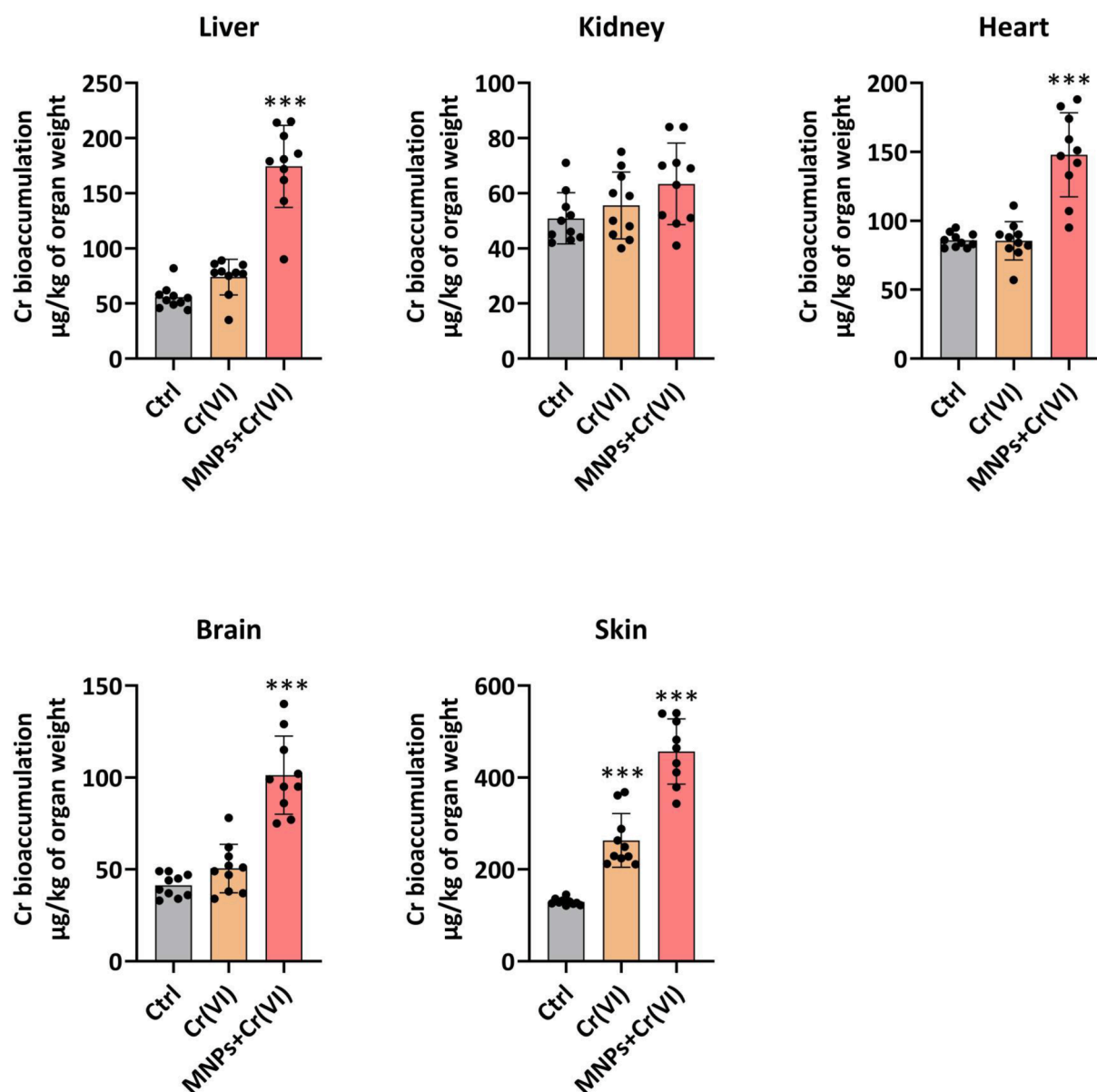


Fig. 3. Chromium bioaccumulation patterns in liver, kidney, heart, brain, and skin of Wistar rats over the 4-week exposure period. The data are reported as the mean $\pm$ standard deviation of 10 rats ($N = 10$) per exposure condition. Statistical significance was assessed by one-way ANOVA followed by the post hoc Dunnett's test. *** $P < 0.001$. Micro-nanoplastics – MNPs; chromium(VI) – Cr(VI).

Altogether, the findings on baseline immunological status revealed an immunosuppressive response under both single and combined exposure, aligning with several previous studies (as reviewed by Rajendran and Chandrasekaran, 2023). Furthermore, a reduced N/L ratio may indicate a shift in immune system dynamics, often reflecting suppression of the innate immune response (decrease in neutrophils) or enhancement of the adaptive immune response (increase in lymphocytes) (Swan and Hickman, 2014; Faria et al., 2016), as observed in the current study. This alteration can be linked to toxicological effects that impair neutrophil function or promote lymphocyte proliferation, often suggesting a reduced capacity to respond to acute stressors or infections. The observed response in the blood, distant from the initial exposure site (i.e., gut epithelia), aligns with recent findings by Shaoyong et al. (2024), who reported that polystyrene particles conditioned with Cr may induce intestinal barrier injury, or so-called gut leakage. They demonstrated that excessive ROS generated by Cr stress damages cell membrane lipids, potentially facilitating the translocation of MNPs into the bloodstream and other tissues. Moreover, since immune cells exhibit

high phagocytic activity, they are likely important targets of MNPs and MNPs + Cr(VI) exposure. As per Shaoyong et al. (2024), autophagy-modulated ROS overproduction within the acidic environment of autophagosomes was shown to accelerate the release of free Cr from polystyrene particles, inducing secondary oxidative stress.

3.5. Alterations in hematological profile following MNPs and Cr(VI) (co-) exposure

In order to evaluate the effects of MNPs and Cr(VI) (co-)exposure on hematological parameters, RBC, Hb, and Hct values, along with RBC indices (MCV, MCH, and MCHC), were evaluated (Fig. 5A). Significantly lower RBC values were observed in all exposed groups compared to the control rats, accompanied by significantly reduced Hb levels, except in the MNPs group, where the Hb values did not change significantly. For the Hct values, no significant variations were found among the exposed groups compared to the control. MCH was significantly increased ($P < 0.05$) in all groups, while MCHC was significantly lower in the Cr(VI)

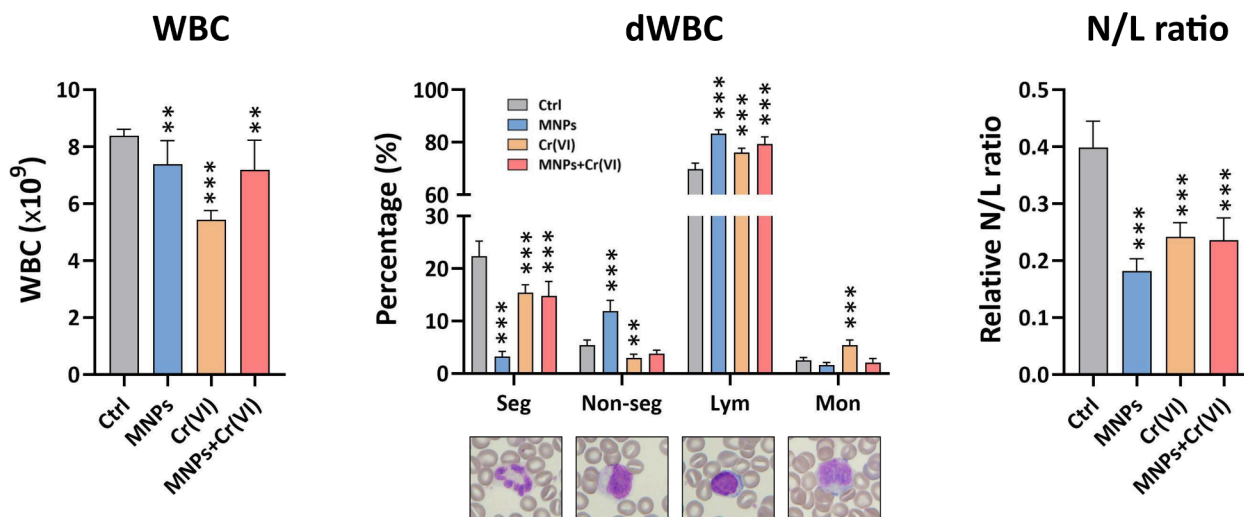


Fig. 4. Micro-nanoplastics and chromium(VI) impact on leukocyte/immune cell counts and neutrophil/lymphocyte ratios following the 4-week exposure in Wistar rats. Left panel: White blood cell count (WBC); Middle panel: differential WBC (dWBC); Right panel: neutrophil/lymphocyte (N/L) ratio. The data are reported as the mean $\pm$ standard deviation of 10 rats ($N = 10$) per exposure condition. Statistical significance was assessed by one-way ANOVA followed by the post hoc Dunnett's test. ** $P < 0.01$; *** $P < 0.001$. Micro-nanoplastics – MNPs; chromium(VI) – Cr(VI); Seg – segmented neutrophils; Non-seg – non-segmented neutrophils; Lym – lymphocytes; Mon – monocytes.

and MNPs + Cr(VI) groups compared to the control. MCV values were significantly lower in the Cr(VI) and MNPs + Cr(VI) groups, except for the MNPs group, which displayed the highest values. In addition, all analyzed parameters were subjected to principal component analysis (PCA), which demonstrated clear separation in the plot for the control, MNPs, and Cr(VI) groups (Fig. 5B). Notably, the MNPs + Cr(VI) group showed more dispersion, clustering closest to the Cr(VI) group but moving slightly towards the control and MNPs groups. The PCA analysis indicated that, in the case of co-exposure, Cr(VI), most probably, is the dominant driver of the observed alterations. Observed changes in hematological indices typically indicate compromised oxygen-carrying capacity of the blood. These alterations are often found associated with conditions such as anemia, reduced erythropoiesis, or hemolysis, and they may reflect underlying toxicity or disruption in RBC production or survival (Rajendran and Chandrasekaran, 2023). Reduced values of RBCs and associated indices have been observed in response to exposure to various MNPs in rats. For example, Yahaya et al. (2024) demonstrated that microplastics elicit reduced levels of RBCs and other hematological indices, consistent with our findings. Furthermore, an additional target of MNPs could be bone marrow, as highlighted by Jing et al. (2022), who found that MNPs disrupt bone marrow cell arrangement, reduce colony-forming, self-renewal, and differentiation capacities, and increase the proportion of lymphocytes, also in agreement with results in Fig. 4. Although this study identified Cr(VI) as the primary driver of toxicity under co-exposure, an interesting finding emerged with MNP exposure alone. Contrary to other results, MCV was found to be increased, which we hypothesize could result from disrupted erythropoiesis due to exposure affecting cell division or as a compensatory mechanism in RBC production.

While Cr(VI) bioaccumulation was significantly higher in most organs under combined exposure, the expected increase in immunotoxicity was not observed (Fig. 4). One possible explanation is that although MNPs facilitated Cr bioaccumulation, they may have also altered its bioavailability, intracellular distribution, and reactivity. MNPs could influence Cr(VI) uptake dynamics by adsorbing Cr onto their surface, potentially modifying its redox state or delaying its cellular interaction, thereby reducing its direct cytotoxic effects on immune and hematopoietic cells. This could lead to lower systemic toxicity despite higher tissue accumulation. Additionally, while Cr(VI) alone is known to induce oxidative stress and immune dysregulation, MNPs have been reported to interact with biological systems through different pathways, such as

modulating gut microbiota (Zhao et al., 2023), or triggering increase in metabolites with antioxidant and anti-inflammatory effects (Song et al., 2024). It is possible that these compensatory responses played a role in reducing some of the immunological disturbances observed with Cr(VI) single exposure. Moreover, different target cells may have varying susceptibility to Cr(VI)-induced toxicity, with immune cells potentially being more resilient under combined exposure due to adaptive responses or altered Cr bioavailability. Regarding the hematological profiles, the absence of a significant difference between combined and single Cr(VI) exposure suggests that Cr(VI) itself was the dominant driver of these alterations, with MNPs primarily affecting Cr distribution rather than its direct hematotoxicity. Future studies focusing on Cr speciation, intracellular Cr interactions, and immune cell-specific responses are needed to clarify these mechanisms.

3.6. Impact of MNPs and Cr(VI) (co-)exposure on serum biomarkers

To evaluate the effects of MNPs and Cr(VI) (co-)exposure on serum parameters as basal biomarkers of organ function and physiological stress, the analysis of serum values is presented in Fig. 6. In this case, more interindividual variability was observed among the exposed groups compared to the control. LDH and CK, as markers of tissue injury, exhibited the greatest individual variations. LDH activity remained unchanged. However, CK activity was found significantly lower ($P < 0.05$) in the MNPs group, suggesting a potential reduction in metabolic or tissue activity, and significantly elevated in the MNPs + Cr(VI) group compared to the control, indicating that combined exposure might have induced tissue damage.

Renal function markers differed between groups, with significantly lower BUN values found in the Cr(VI) group and significantly higher creatinine values in the MNPs group compared to the control. These findings partially align with Wang et al. (2023), who reported that exposure to the same concentration of polystyrene microparticles ($1 \mu\text{m}$) for the same duration resulted in increases in both BUN and serum creatinine levels.

Liver markers (AST, ALT, and Bil) showed considerable variability among individual animals, but these changes were not statistically significant. Glucose values appeared to vary the least between groups, with no significant differences observed staying in agreement with Liu et al. (2022). Protein and albumin levels in the exposed groups were similar to those in the control group; however, globulin levels under each exposure

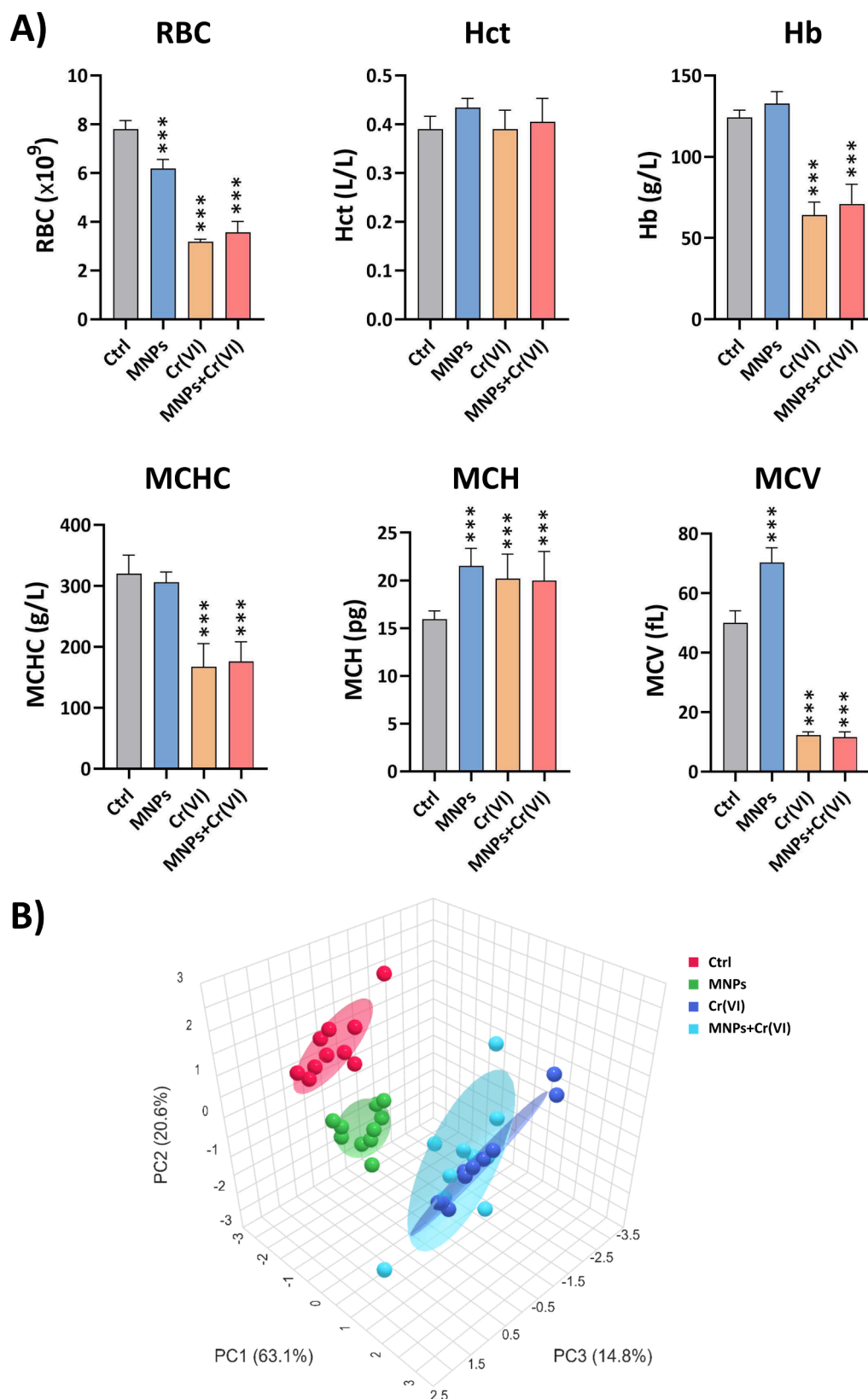


Fig. 5. Micro-nanoplastics and chromium(VI) impact on red blood cell indices following the 4-week exposure in Wistar rats. (A) RBC – red blood cells; Hct – hematocrit; Hb – hemoglobin; MCV – mean corpuscular volume; MCH – mean corpuscular hemoglobin; MCHC – mean cell corpuscular hemoglobin. The data are reported as the mean $\pm$ standard deviation of 10 rats ($N = 10$) per exposure condition. Statistical significance was assessed by one-way ANOVA followed by the post hoc Dunnett's test. *** $P < 0.001$. (B) Principal component analysis (PCA) of the red blood cell indices for different exposure conditions. Micro-nanoplastics – MNPs; chromium(VI) – Cr(VI). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

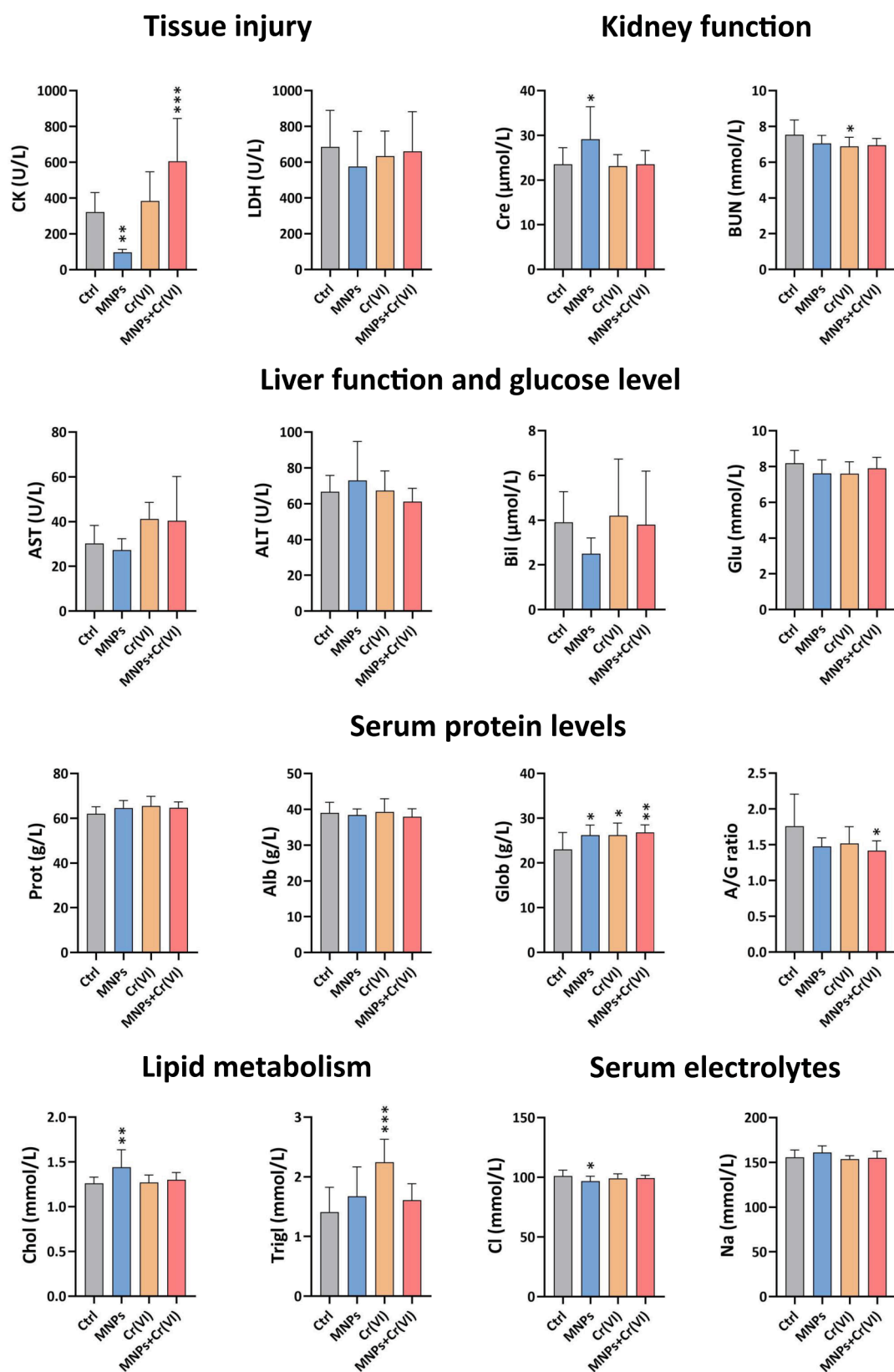


Fig. 6. Micro-nanoplastics and chromium (VI) impact on serum biomarkers following the 4-week exposure in Wistar rats. CK – creatine kinase; LDH – lactate dehydrogenase; Cre – creatinine; BUN – blood urea nitrogen; AST – aspartate aminotransferase; ALT – alanine aminotransferase; Bil – bilirubin; Glu – glucose; Prot – proteins; Alb – albumins; Glob – globulins; A/G ratio – albumin/globulin ratio; Chol – cholesterol; Trigl – triglycerides. The data are reported as the mean $\pm$ standard deviation of 10 rats (N = 10) per exposure condition. Statistical significance was assessed by one-way ANOVA followed by the post hoc Dunnett's test. *P < 0.05; **P < 0.01; ***P < 0.001. Micro-nanoplastics – MNPs; chromium(VI) – Cr(VI).

scenario were higher and significantly different ($P < 0.05$) compared to the control. Consequently, the A/G ratio was found slightly lower in all exposed groups, with a significant difference noted only in the case of MNPs + Cr(VI). Serum electrolytes (Cl^- and Na^+) did not exhibit large variations, except for significantly lower chloride levels in the MNPs group.

Regarding the lipid markers in serum, cholesterol and triglyceride levels were assessed. While cholesterol levels in the Cr(VI) and MNPs + Cr(VI) groups were comparable to those in the control group, significantly higher values were observed in the MNPs group. Several studies have reported alterations in lipid metabolism following exposure to polystyrene particles in rats and mice. For instance, higher cholesterol levels were noted; however, these effects appear to be influenced by various factors, including maternal exposure, sex, particle size, and other variables (Luo et al., 2019; Nnoruka et al., 2022; Yahaya et al., 2024). In contrast, significantly higher triglyceride levels were found in the Cr(VI) group compared to the control. Even if limited, the observed changes in serum of rats indicate potential multi-level physiological stress reflected by alterations in serum biomarkers both under single and combined exposure.

The apparent attenuation of some MNP-induced effects under combined exposure may be due to Cr(VI) triggering adaptive physiological responses, influencing detoxification pathways, oxidative stress regulation, or metabolic adjustments. Additionally, interactions between MNPs and Cr(VI) may have altered their bioavailability and toxicokinetics, affecting their systemic distribution and cellular interactions. Inter-individual variability among the animals could also contribute to fluctuations in biomarker levels. Importantly, the current findings represent a snapshot after four weeks of exposure, and it remains unknown how these responses may have differed at earlier or later time points. Future studies incorporating multiple exposure durations will be essential to further elucidate these temporal dynamics.

4. Conclusions

This study highlighted the interaction between micro- and nano-sized polystyrene particles and their effects when co-exposed with Cr(VI) in rats. Nano-sized particles were observed co-aggregating on the surface of micro-sized particles, significantly increasing the surface area and potentially influencing adverse effects. Under Cr(VI) and MNP co-exposure, significantly higher bioaccumulation of Cr was found in organs such as the liver, heart, and brain, suggesting that MNPs enhanced Cr absorption and tissue distribution. Co-exposure also notably reduced WBC and segmented neutrophil counts, with elevated lymphocyte levels in the MNP-exposed group. Hematological profiles showed reduced RBC counts and Hb levels across exposure groups, with PCA revealing Cr(VI) as the dominant driver of these alterations. Serum biomarkers indicated physiological stress, with elevated creatinine and cholesterol levels in the MNP group and higher triglycerides in the Cr(VI) group. In summary, polystyrene MNPs increase Cr(VI) distribution and bioavailability, but co-exposure does not uniformly exacerbate toxicity. Rather, their interaction may lead to selective physiological alterations, emphasizing the need to better understand their combined effects and potential health risks.

CRedit authorship contribution statement

Damir Suljević: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Patrik Karlsson:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Formal analysis. **Muhamed Foćak:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis. **Maja Mitrašinić Brulić:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis. **Jasmina Sulejmanović:** Writing – review & editing, Writing –

original draft, Validation, Resources, Methodology, Investigation, Formal analysis. **Elma Šehović:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis. **Eva Särndahl:** Writing – review & editing, Validation, Supervision, Project administration, Funding acquisition. **Magnus Engwall:** Writing – review & editing, Validation, Supervision, Project administration, Funding acquisition. **Andi Alijagić:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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